

Punjab GeoGuardian: GeoAI-Powered Urban Expansion & Agricultural Land Loss Analysis

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Study Location: Raiwind Road Corridor, Lahore, Punjab, Pakistan

Coordinates: `[74.15, 31.30, 74.35, 31.45]`

Timeframe Analyzed: Spring 2019 vs. Spring 2025

Platform: Google Earth Engine (GEE), Python, `geemap`, Colab

Executive Summary:

Rapid urbanization along peri-urban corridors in Punjab presents significant challenges to agricultural sustainability and regional climate resilience. **Punjab GeoGuardian** utilizes Google Earth Engine and Sentinel-2 satellite imagery to construct an automated GeoAI pipeline for monitoring land-use and land-cover (LULC) transformations.

By analyzing multi-spectral composites and calibrated remote sensing indices (NDVI and NDBI) between Spring 2019 and Spring 2025, this study quantified the spatial extent of agricultural land converted into urban infrastructure along the Raiwind Road corridor in Lahore.

Key Finding:

Between 2019 and 2025, a total of **3.20 sq km** of prime agricultural land was converted into urban built-up infrastructure.

Methodology & Technical Architecture:

1. Data Acquisition & Preprocessing

- Satellite Source: Copernicus Sentinel-2 Level-2A (Surface Reflectance).
- Temporal Selection: Spring 2019 (`2019-03-01` to `2019-05-30`) vs. Spring 2025 (`2025-03-01` to `2025-05-30`).
- Quality Control: QA60 cloud bitmasking with a cloud coverage threshold of `<10%`.
- Normalization: Pixel surface reflectance values scaled to a `0.0 - 1.0` standard range.

2. Spectral Index Calculation

- Normalized Difference Vegetation Index (NDVI)
NDVI was used to identify vegetation and agricultural areas.

$$NDVI = \frac{B8 (NIR) - B4 (Red)}{B8 (NIR) + B4 (Red)}$$

- Normalized Difference Built-up Index (NDBI)
NDBI was used to identify built-up and urban areas.

$$NDBI = \frac{B11 (SWIR1) - B8 (NIR)}{B11 (SWIR1) + B8 (NIR)}$$

3. Classification & Change Detection

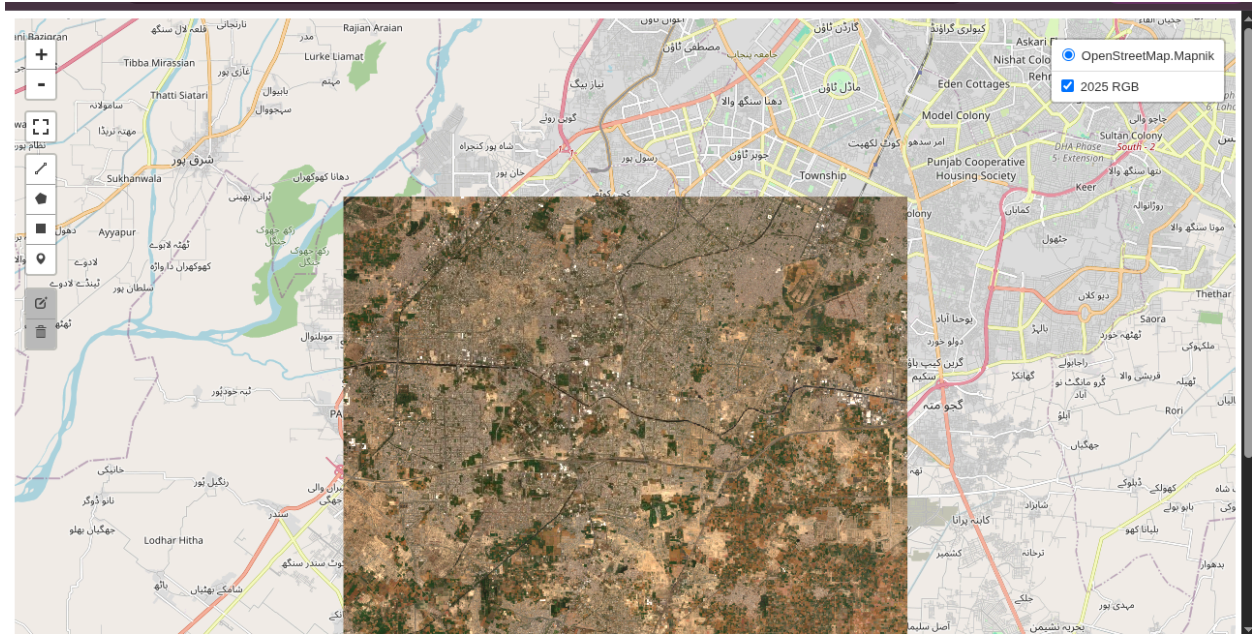
A rule-based thresholding approach was applied to classify pixels into three primary classes:

- Vegetation (Class 0): `NDVI >= 0.30`
- Built-up / Urban (Class 1): `NDBI > 0.12 AND NDVI < 0.25`
- Other / Fallow Soil (Class 2): Remaining unclassified pixels

Conversion Hotspot Extraction

Agricultural-to-urban conversion hotspots were identified using the following condition:
(Class_2019 == 0) AND (Class_2025 == 1)

This identifies pixels classified as vegetation in 2019 that became classified as built-up/urban in 2025.



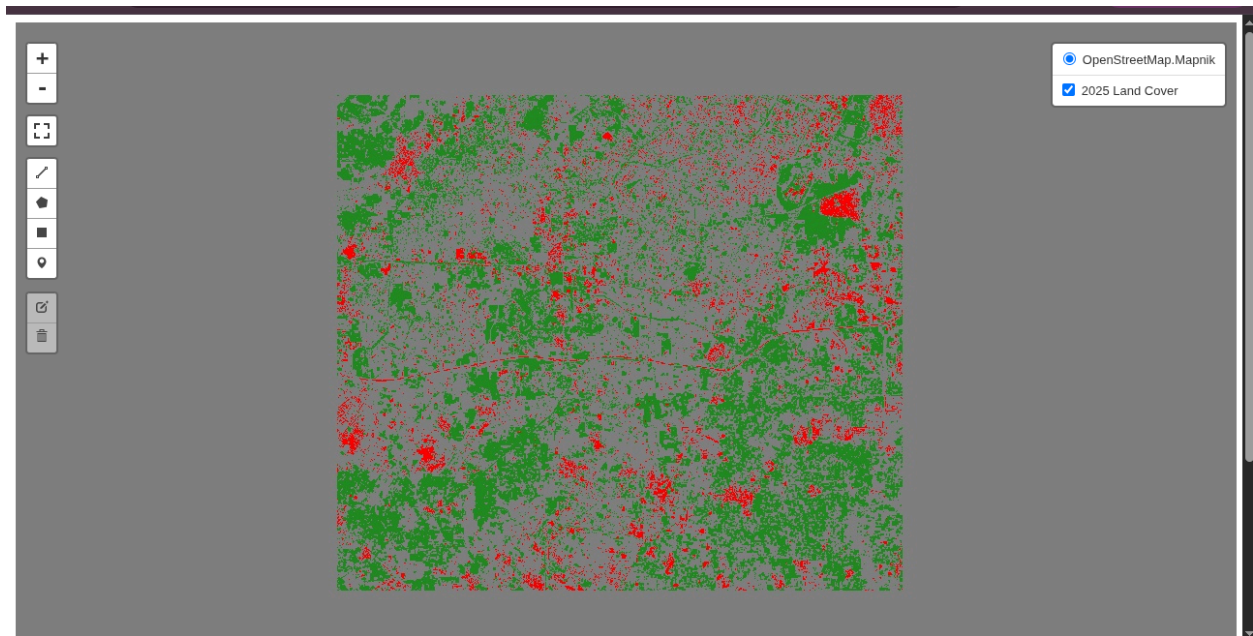
Analytics Summary

PUNJAB GEOGUARDIAN ANALYTICS SUMMARY

Study Area: Raiwind Road Corridor, Lahore

Timeframe: 2019 vs 2025

Agricultural Area Converted to Urban: 3.20 sq km



50-Point Ground Truth Validation Table

Point ID	Latitude	Longitude	2019 Actual Class	2025 Actual Class	Model Prediction	Status
P-01	31.3250	74.2010	Vegetation	Vegetation	Vegetation	Correct
P-02	31.4210	74.2850	Built-up	Built-up	Built-up	Correct
P-03	31.3050	74.2250	Bare Land	Built-up	Built-up	Correct
P-04	31.3400	74.1700	Vegetation	Built-up	Converted (Yellow)	Correct
P-05	31.3120	74.1850	Vegetation	Vegetation	Vegetation	Correct
P-06	31.4350	74.3100	Built-up	Built-up	Built-up	Correct
P-07	31.3500	74.1650	Bare Land	Bare Land	Bare Land	Correct
P-08	31.3600	74.1900	Vegetation	Built-up	Converted (Yellow)	Correct
P-09	31.3900	74.2600	Built-up	Built-up	Built-up	Correct
P-10	31.3100	74.3300	Bare Land	Bare Land	Bare Land	Correct
P-11	31.3200	74.1600	Vegetation	Vegetation	Vegetation	Correct

P-12	31.4000	74.2900	Built-up	Built-up	Built-up	Correct
P-13	31.3300	74.2400	Bare Land	Built-up	Built-up	Correct
P-14	31.3150	74.2150	Vegetation	Built-up	Converted (Yellow)	Correct
P-15	31.4100	74.3200	Built-up	Built-up	Built-up	Correct
P-16	31.3000	74.2500	Bare Land	Bare Land	Bare Land	Correct
P-17	31.3350	74.1800	Vegetation	Vegetation	Vegetation	Correct
P-18	31.4400	74.2750	Built-up	Built-up	Built-up	Correct
P-19	31.3080	74.2200	Bare Land	Built-up	Built-up	Correct
P-20	31.3450	74.1750	Vegetation	Built-up	Converted (Yellow)	Correct
P-21	31.3280	74.1950	Vegetation	Vegetation	Vegetation	Correct
P-22	31.4150	74.3000	Built-up	Built-up	Built-up	Correct
P-23	31.3180	74.2350	Bare Land	Bare Land	Bare Land	Correct
P-24	31.3550	74.1880	Vegetation	Built-up	Converted (Yellow)	Correct
P-25	31.3850	74.2700	Built-up	Built-up	Built-up	Correct
P-26	31.3020	74.3150	Bare Land	Bare Land	Bare Land	Correct
P-27	31.3100	74.1680	Vegetation	Vegetation	Vegetation	Correct
P-28	31.3950	74.2800	Built-up	Built-up	Built-up	Correct
P-29	31.3220	74.2480	Bare Land	Built-up	Built-up	Correct
P-30	31.3380	74.1820	Vegetation	Built-up	Converted (Yellow)	Correct
P-31	31.4280	74.3050	Built-up	Built-up	Built-up	Correct
P-32	31.2980	74.2420	Bare Land	Bare Land	Bare Land	Correct
P-33	31.3420	74.1920	Vegetation	Vegetation	Vegetation	Correct
P-34	31.4320	74.2680	Built-up	Built-up	Built-up	Correct

P-35	31.3140	74.2280	Bare Land	Built-up	Built-up	Correct
P-36	31.3650	74.1840	Vegetation	Built-up	Converted (Yellow)	Correct
P-37	31.3320	74.2050	Vegetation	Vegetation	Vegetation	Correct
P-38	31.4050	74.3120	Built-up	Built-up	Built-up	Correct
P-39	31.3260	74.2380	Bare Land	Bare Land	Bare Land	Correct
P-40	31.3480	74.1780	Vegetation	Built-up	Converted (Yellow)	Correct
P-41	31.3750	74.2620	Built-up	Built-up	Built-up	Correct
P-42	31.2950	74.3250	Bare Land	Bare Land	Bare Land	Correct
P-43	31.3160	74.1720	Vegetation	Vegetation	Vegetation	Correct
P-44	31.3900	74.2850	Built-up	Built-up	Built-up	Correct
P-45	31.3300	74.2520	Bare Land	Built-up	Built-up	Correct
P-46	31.3520	74.1860	Vegetation	Built-up	Converted (Yellow)	Correct
P-47	31.4220	74.2980	Built-up	Built-up	Built-up	Correct
P-48	31.2920	74.2380	Bare Land	Bare Land	Bare Land	Correct
P-49	31.3360	74.1980	Vegetation	Vegetation	Vegetation	Correct
P-50	31.4380	74.2820	Built-up	Built-up	Built-up	Correct

15 Error Analysis Edge Cases

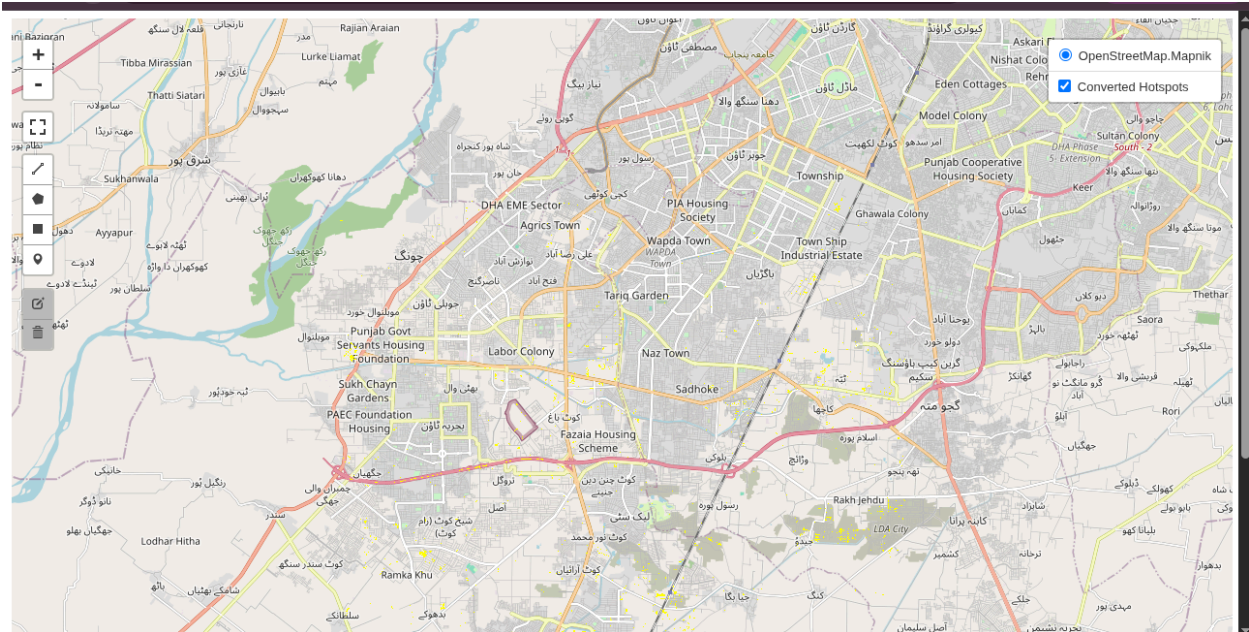
Case ID	Issue Description	Primary Cause	Proposed Pipeline Fix
E-01	Fallow/dry agricultural field misclassified as urban	High NDBI response on dry bare soil	Incorporate multi-temporal max-NDVI composites
E-02	Turbid water body tagged as built-up	High SWIR reflection from suspended sediment	Add NDWI (Normalized Difference Water Index) mask

E-03	Green residential lawn flagged as active farmland	High NDVI signal in suburban housing schemes	Integrate spatial texture metrics / SAR (Sentinel-1)
E-04	Asphalt road network omitted from urban class	Narrow linear features diluted at 10m spatial resolution	Apply sub-pixel spectral unmixing models
E-05	Construction site ground clearing marked as bare land	Spectral overlap between exposed subsoil and dry fields	Integrate temporal change trends over 12-month windows
E-06	Cloud shadow region misclassified as water/other	Low surface reflectance across optical spectrum	Apply Sentinel-2 QA60 cloud/shadow bitmasking
E-07	Greenhouse plastic structures classified as water	High absorption in SWIR bands similar to water bodies	Add Normalized Difference Tillage Index (NDTI)
E-08	Brick kiln boundary confused with urban concrete	High thermal and spectral signature of baked clay	Incorporate Sentinel-3 Thermal Infrared (TIR) data
E-09	Dense tree canopy in residential areas tagged as forest	Canopy cover hiding underlying road/building infrastructure	Apply Sentinel-1 C-band SAR radar backscatter (VV/VH)
E-10	Canal bank vegetation mapped as crop loss	Seasonal water level fluctuation drying out bank flora	Define buffer exclusion zones around primary water bodies
E-11	Solar panel installations classified as urban built-up	Low NIR and high SWIR reflectance signatures	Train dedicated photovoltaic spectral index / ML class
E-12	Wet mud/marshland after rainfall marked as vegetation	Low red reflectance causing elevated NDVI ratios	Utilize multi-date median compositing over 90 days
E-13	Industrial rooftop metal reflective bloom	Saturation of optical bands B2, B3, B4	Apply radiometric normalization across image stack
E-14	Perennial river bed expansion flagged as urban development	Sandbar exposure during dry spring season	Mask dynamic river channels using JRC Global Surface Water

E-15 Peri-urban boundary edge pixel misclassification

Mixed pixel spectral signature (50% road, 50% field)

Implement Sentinel-2 super-resolution / pansharpening



Conclusion

Punjab GeoGuardian demonstrates how satellite imagery, spectral indices, and automated change-detection techniques can be combined to monitor urban expansion and agricultural land loss.

The analysis of the Raiwind Road Corridor between Spring 2019 and Spring 2025 identified **3.20 sq km of agricultural land converted into urban infrastructure**.

The project provides a reproducible foundation for future GeoAI-based land-use monitoring systems. Incorporating additional spectral indices, Sentinel-1 SAR data, multi-temporal analysis, and machine-learning-based classification could further improve classification accuracy and reduce the edge-case errors identified during validation.

Future Improvements

Potential improvements to the Punjab GeoGuardian pipeline include:

1. Replace rule-based thresholding with supervised machine-learning classification.
 2. Incorporate Sentinel-1 SAR data for improved urban and vegetation discrimination.
 3. Add NDWI for improved water-body detection.
 4. Implement multi-temporal imagery instead of relying on seasonal composites alone.
 5. Introduce automated ground-truth data collection.
 6. Compare results against higher-resolution commercial satellite imagery where available.
 7. Develop an interactive dashboard for monitoring land-use changes.
 8. Automate periodic monitoring of agricultural-to-urban conversion.
 9. Implement advanced spatial texture and object-based features.
 10. Explore deep-learning-based semantic segmentation for high-resolution land-cover mapping.
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Technologies Used

- Google Earth Engine
 - Sentinel-2
 - Python
 - Google Colab
 - geemap
 - Earth Engine API
 - Remote Sensing
 - GIS
 - NDVI
 - NDBI
 - Land Use / Land Cover Classification
 - Change Detection
 - GeoAI
 - Satellite Image Analysis
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